Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/27/2007
PI Name:	Eckmann, David M. Ph.D., M.D.		
Project Title:	Macromolecule Adsorption and Bubble Adhesion to Model Endothelial Surface Layers		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Biofluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104 - 6802	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	01/01/2005	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: grant reinstated 8/06; end date changed to 9/30/2007, per F. Chiaramonte (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ayyaswamy, Portonovo (University of Pennsylvania) Composto, Russell (University of Pennsylvania)		
Grant/Contract No.:	NNC05GA30G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The pathophysiology of gas embolism disease has been studied little and is poorly understood. The initiating events involve bubble adhesion within the vasculature. The molecular mechanical basis of bubble adhesion to the vessel wall causing blood flow obstruction is unknown. Our preliminary experiments performed in vivo indicate that blood flow obstruction is caused by adhesion of the bubble surface to the endothelial glycocalyx, but the details of adhesive interactions remain undetermined. Our global hypothesis is that adhesion interactions between the molecular entities present in the endothelial glycocalyx and resident on the bubble-blood interface are non specific (i.e., not ligand-receptor mediated), but depend on surface adsorption of circulating blood-borne macromoles to the gas-liquid interface. The subsequent development of adhesion interactions between adsorbed macromolecules and elements of the endothelial surface layer then controls bubble lodging. To identify determinants of bubble adhesion to the vessel wall, we will use in vitro methods including adhesion force measurement of microbubbles (or microdrops) of biofluids (e.g., protein solutions, plasma) to biomimetic glycocalyx constructs coating capillary tubes and flat surfaces. The results of experiments will be compared to measurements made in ongoing experiments using excised arteriolar blood vessels. Alterations of pH will be used to control electrostatic interactions. Once the major molecular determinants of adhesion are identified, we predict that modulation of adhesion by introduction of surfactants targeted to bubble-biofluid (blood) interfaces will reduce net adhesion force. The experiments will be coupled with computational modeling of spreading and peeling dynamics of bubble adhesion and flow obstruction in a flexible-walled vessel. The influence of microgravity will be incorporated into the numerical modeling to reduce bubble buoyancy and seek nonaxisymmetric and nonuniform adhesion profiles, which cannot be elicited in ground-based experiments. The numerical experiments will be used to develop future experiments for space-based testing. Outreach and education programs will include research opportunities for undergraduate students, high school science teachers and high school students and seminars open to the faculty and students of local universities and high schools. In a broad application, this work addresses interfacial fluid mechanics and molecular mechanics of adhesion and adsorption as well as the materials science of surface coating development. The more specific physiological implication of this work is the development of a potential therapy or intervention which will preserve end-organ flood flow and minimize development of injury in embolized tissue and organs, particularly in an environment in which no other therapy is accessible. This is particularly important to NASA: its Bioastronautics Critical Path Roadmap (CPR) Risk #47 clearly identifies gas embolization resulting from decompression during space flight to be one of the 55 most critical risks to human safety, health, and performance to be expected from mid- and far-term long duration human space flight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work investigates specifically the elements of adhesion interactions between surface adsorbed blood-borne macromolecules on bubble surfaces and the endothelial glycocalyx. This has impact on Earth for application to decompression illness and gas embolism resulting from cardiopulmonary bypass operations.
Task Progress:	In the truncated grant period we completed experiments on bubble adhesion to coated microtubes. Major findings are that dextran grafting density and dextran molecular weight are major determinants of adhesion force. This work is currently in revision for peer reviewed publication (Ombelli M, Eckmann DM and Composto RJ. Characterization of surface coated materials for the in vitro study of microvascular gas embolism.J Biomed Mat Res.). We have completed computational work, which has been presented at international conferences (Interdisciplinary Transport Phenomena in Microgravity and Space Sciences IV, Tomar, Portugal, August, 2005; Mukundakrishnan, K., Quan, S., Eckmann, D.M., Ayyaswamy, P.S.: Bubble motion through a generalized power-law fluid in a vertical tube. Interdisciplinary Transport Phenomena in Microgravity and Space Sciences V, Bansko, Bulgaria October 2007). Three papers have been published (Ayyaswamy PS, Zhang J, and Eckmann DM. Numerical modeling of the transport to an intravascular bubble in a tube with a soluble/insoluble surfactant. Ann NY Acad Sci, 1077: 270-287, 2006; Eckmann, D.M., Zhang, J., Ayyaswamy, P.S.: Surfactant based intervention to eliminate gas embolism during long duration space based activity. Ann NY Acad Sci, 1077: 256-269, 2006; Mukundakrishnan K, Quan S, Eckmann DM and Ayyaswamy PS. Buoyant bubble rise in a finite, liquid-filled circular cylinder. Phl Rev E, 76:036308 1-15, 2007) and two more are in press (Mukundakrishnan K, Quan S, Eckmann DM and Ayyaswamy PS: Drop motion through a Carreau-Yasuda fluid flowing in a vertical tube. Accepted to Proceedings, ISHMT-ASME Heat and Mass Transfer Conference, Hyderabad, India, January, 2008; Mukundakrishnan K, Eckmann DM and Ayyaswamy PS. Bubble motion through a generalized power-law fluid flowing in a vertical tube. In Press, Ann NY Acad Sci, 2008). In this work we have used a newly developed algorithm in conjunction with the front tracking scheme, we have evaluated the transport associated with a deformable bubble moving in a tube in the presence of a soluble or an insoluble surfactant. The numerical evaluations indicate that as the location of the adsorptive interface gets closer to the vessel wall, the surfactant amount on the wall gets depleted. The implication is that the process by which a bubble occluding a vessel dislodges may depend both on the strength of the diffusivity of the surfactant and the adsorption process. Elements of this computational work demonstrates how bulk and surface transport of a surfactant provide competition for interfacial occupancy such that surfactant adsorption can dominate protein (or other macromolecules) adsorption and also aid in protein being blocked or expelled from the interface. We demonstrate that the presence of surfactant along the gas-liquid interface also induces variation in the interfacial tension, which in turn affects the blood flow and the bubble motion. The computational work described has been performed in part using an 8-node cluster of workstations purchased in part with equipment funds provided by the grant.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Eckmann DM, Zhang J, Lampe J, Ayyaswamy PS. "Gas embolism and surfactant-based intervention: implications for long-duration space-based activity." Ann N Y Acad Sci. 2006 Sep;1077:256-69. Review. PMID: 17124129 , Sep-2006
Articles in Peer-reviewed Journals	Ayyaswamy PS, Zhang J, Eckmann DM. "Numerical modeling of the transport to an intravascular bubble in a tube with a soluble/insoluble surfactant." Ann N Y Acad Sci. 2006 Sep;1077:270-87. PMID: 17124130 , Sep-2006

Articles in Peer-reviewed Journals	Mukundakrishnan K, Quan S, Eckmann DM, Ayyaswamy PS. "Numerical study of wall effects on buoyant gas-bubble rise in a liquid-filled finite cylinder." Phys Rev E Stat Nonlin Soft Matter Phys. 2007 Sep;76(3 Pt 2):036308. PMID: 17930342 , Sep-2007
Articles in Peer-reviewed Journals	Mukundakrishnan K, Eckmann DM, Ayyaswamy PS. "Bubble motion through a generalized power-law fluid flowing in a vertical tube." Ann N Y Acad Sci. 2009 Apr;1161:256-67. PubMed PMID: 19426324 , Apr-2009
Articles in Peer-reviewed Journals	Ombelli M, Eckmann DM, Composto RJ. "Characterization of surface coated materials for the in vitro study of microvascular gas embolism." Journal of Biomedical Materials Research, in revision, December 2007. , Dec-2007
Papers from Meeting Proceedings	Mukundakrishnan K, Quan S, Eckmann DM, Ayyaswamy PS. "Drop motion through a Carreau-Yasuda fluid flowing in a vertical tube." 19th National & 8th ISHMT-ASME Heat and Mass Transfer Conference, Hyderabad, India, January 3-5, 2008. Accepted for Proceedings, 19th National & 8th ISHMT-ASME Heat and Mass Transfer Conference, in press, December 2007. , Dec-2007